Portfolio Website PRD

Oct 12, 2026 · @salman siddik shovo

1. Overview

A single-page personal portfolio for Salman Siddiq Shovo that lets a recruiter or client see his work and contact him within one minute.

Goals

- Present Salman as a junior frontend developer (CSE student, Daffodil International University).
- Show real projects, pulled live from GitHub (`salmansiddikshovo`).
- Give visitors one clear action: view work, download the CV, or send a message.
- Prove skills with the stack itself: Next.js, TanStack Query, Axios, shadcn/ui.

Target audience

- Recruiters and hiring managers (first look at Hero and Projects, usually under 30 seconds).
- Freelance clients and collaborators.
- Teachers, peers and the open-source community.

Success metrics

Metric	Target
Lighthouse (Performance, Accessibility, SEO, Best Practices)	90+ each
Largest Contentful Paint	under 2.5 s on mobile
Contact form submit success rate	95%+
Time to find projects from the top of the page	under 10 s

2. Scope

Version 1 is one scrolling home page with seven sections, one API route and no database.

In scope (v1)

- Home page with: Hero, About, Skills, Projects, Experience, Education, Contact, plus Navbar and Footer.
- Dark and light theme.
- Projects loaded from the GitHub REST API.
- Contact form that sends an email through `/api/contact`.
- Downloadable CV (`/resume.pdf`).
- SEO metadata and Open Graph image.
- Fully responsive layout (mobile first).

Out of scope (v1)

- User accounts, login or admin dashboard.
- Blog, CMS and database.
- Multi-language switching (English only for the UI).
- Analytics dashboards (a simple Vercel Analytics script is optional).

Later (v2 ideas)

- Testimonials, Services and Certifications sections.
- Blog with MDX or a headless CMS.
- Project detail pages at `/projects/[slug]`.
- Bangla language toggle.

3. Tech stack

The stack is Next.js App Router with TypeScript, styled by Tailwind and shadcn/ui, with Axios and TanStack Query handling all remote data.

Tool	Role	Rule
Next.js (App Router)	Routing, SEO metadata, API route	Server components by default; add `"use client"` only when hooks or events are needed
TypeScript	Type safety	`strict: true`; no `any`
Tailwind CSS	Styling	Use theme tokens (`bg-background`, `text-muted-foreground`), no hard-coded hex colors
shadcn/ui	UI components	Add components with the CLI; never rewrite them from scratch

Tool	Role	Rule
Axios	HTTP client	Only through the instances in `lib/axios.ts`
TanStack Query	Server state, caching, mutations	All fetching goes through hooks in `hooks/`
next-themes	Dark and light mode	Class-based theme, system default
react-hook-form + zod	Form state and validation	One zod schema per form
sonner	Toast messages	Success and error feedback only
react-icons, lucide-react	Icons	Brand icons from react-icons, UI icons from lucide
Resend or Nodemailer	Sending contact emails	Called only inside the API route

Install commands

```
npx create-next-app@latest portfolio --typescript --tailwind --app --src-dir
npx shadcn@latest init
npx shadcn@latest add button card badge avatar sheet separator skeleton form
input textarea sonner
npm i axios @tanstack/react-query next-themes react-hook-form zod
@hookform/resolvers react-icons lucide-react
```

Environment variables

Variable	Purpose
`NEXT_PUBLIC_API_URL`	Optional base URL for Axios (defaults to `/api`)
`RESEND_API_KEY`	Server-only key for sending email
`CONTACT_TO_EMAIL`	Address that receives messages
`NEXT_PUBLIC_SITE_URL`	Used for canonical URL and Open Graph

4. Content and personal data

All personal content lives in one file, `src/data/profile.ts`, so no component contains hard-coded personal text.

Profile

Field	Value
Name	Salman Siddiq Shovo
Role shown in Hero	Frontend Developer
University	B.Sc. in CSE, Daffodil International University
HSC	2020, Milestone College, GPA 4.87
SSC	2018, Sristy Academic College, GPA 4.61
Languages	Bangla, English
Location	Dhaka, Bangladesh

Social links

Platform	URL
Facebook	salmansiddik.shovo
Instagram	salman_siddik_shovo_official
LinkedIn	salman-shovo-a307912a1
GitHub	salmansiddikshovo

Content rules

- Write the bio in first person, 2 to 4 sentences, plain English.
- Hero tagline: one sentence, under 25 words.
- List only skills Salman can talk about in an interview.
- Never invent experience, projects, awards or numbers.
- Every external link opens in a new tab with `rel="noopener noreferrer"`.

Assets needed

- `public/me.jpg`: square profile photo, at least 600 x 600 px, under 300 KB.
- `public/resume.pdf`: current CV, under 1 MB.
- `public/og.png`: Open Graph image, 1200 x 630 px.

5. Page sections

Each section is one component in `components/sections/` wrapped in `SectionWrapper`, with an `id` the Navbar links to. Order on the page is top to bottom as listed.

#	Section	Must show	shadcn/ui parts	Data source
1	Navbar	Logo, 6 anchor links, theme toggle, "Hire me" button, mobile menu	`Button`, `Sheet`	Local array
2	Hero	Name, role, tagline, "View my work" and "Download CV" buttons, social icons, photo	`Avatar`, `Badge`, `Button`	`profile.ts`
3	About	Short bio, location, languages	`Card`, `Badge`	`profile.ts`
4	Skills	Skills grouped by category	`Card`, `Badge`	`profile.ts`
5	Projects	Up to 6 repos: name, description, language, stars, Code and Live links	`Card`, `Badge`, `Button`, `Skeleton`	GitHub API (TanStack Query)
6	Experience	Timeline of roles with achievement bullets	`Card`	`profile.ts`
7	Education	Degree, institution, year, GPA	`Card`, `Badge`	`profile.ts`
8	Contact	Name, email, message form, email address, social icons	`Form`, `Input`, `Textarea`, `Button`, `sonner`	`/api/contact` mutation
9	Footer	Copyright year, social icons	`Separator`	`profile.ts`

Behaviour rules

- Navbar is sticky with a blurred background; the mobile menu opens in a `Sheet` and closes after a link is tapped.
- Anchor links scroll smoothly and leave room for the sticky Navbar (`scroll-mt-20`).
- Projects must show 3 skeleton cards while loading, an error message with a "Try again" button on failure, and a friendly empty state when there are no repos.

- Projects hides forks and sorts by stars, highest first.
- Experience renders nothing when the list is empty, so no blank heading appears.
- Contact disables the submit button while sending, clears the form on success and shows a toast for success or failure.
- The Hero photo falls back to the initials "SS" if `me.jpg` is missing.

Acceptance criteria

- [ ] Every section is reachable from the Navbar on desktop and mobile.
- [ ] All 4 social links open the correct profiles in a new tab.
- [ ] CV downloads from the Hero button.
- [ ] Projects shows real GitHub data and survives an API failure without crashing.
- [ ] Contact form rejects an invalid email and a message under 10 characters.

6. Data and API rules

Components never call Axios directly: they use a hook, and the hook uses an Axios instance.

Axios

- Two instances in `src/lib/axios.ts`: `api` (own backend, base URL `/api`) and `githubApi` (`https://api.github.com`).
- Timeout is 15 seconds on both.
- The `api` response interceptor turns any failure into an `Error` with a readable message.
- Never put secrets in client code; anything secret runs inside the API route.

TanStack Query

- One `QueryClient`, created once inside `QueryProvider` with `useState`.
- Defaults: `staleTime` 5 minutes, `refetchOnWindowFocus` off, `retry` 1.
- Query keys are arrays, most general first, and never typed twice in different files.
- Reads use `useQuery`; writes use `useMutation`.
- Every query component handles four states: loading, error, empty and success.

Hook	Type	Query key	Endpoint
`useGithubRepos(username)`	`useQuery`	`["github-repos", username]`	`GET /users/{username}/repos`

Hook	Type	Query key	Endpoint
`useContact()`	`useMutation`	none	`POST /api/contact`

Contact API route (`app/api/contact/route.ts`)

Case	Response
Valid body	`200` with `{ "ok": true }` after sending the email
Invalid body (zod fails)	`400` with `{ "message": "Invalid form data" }`
Email provider fails	`500` with a generic message, details only in server logs

- The same zod rules apply on client and server: name 2+ characters, valid email, message 10+ characters.
- Add basic spam protection before launch: a hidden honeypot field, and optionally rate limiting by IP.
- GitHub allows 60 unauthenticated requests per hour per IP. TanStack Query caching keeps visitors well under this; if it ever fails, the UI shows the error state and nothing else breaks.

7. Project structure

Files are grouped by job: pages in `app/`, sections in `components/sections/`, data access in `hooks/`, and personal content in `data/`.

```
src/
├── app/
│   ├── layout.tsx            # fonts, metadata, providers, Toaster
│   ├── page.tsx              # assembles all sections
│   ├── globals.css
│   └── api/contact/route.ts  # POST: validate + send email
├── components/
│   ├── ui/                   # shadcn components (CLI-generated)
│   ├── sections/             # Hero, About, Skills, Projects,
│   │                         # Experience, Education, Contact
│   ├── Navbar.tsx
│   ├── Footer.tsx
│   ├── ThemeToggle.tsx
│   ├── SocialLinks.tsx
│   └── SectionWrapper.tsx
├── hooks/
```

```
|   ├── useGithubRepos.ts
|   └── useContact.ts
├── lib/
|   ├── axios.ts              # api + githubApi instances
|   └── utils.ts              # shadcn cn() helper
├── providers/
|   └── QueryProvider.tsx
├── data/
|   └── profile.ts            # all personal content
└── types/
    └── index.ts              # shared interfaces
public/
├── me.jpg
├── resume.pdf
└── og.png
```

Architecture rules

- Dependencies point one way: `sections` use `hooks`, `hooks` use `lib`, and `lib` uses nothing from the app.
- `app/page.tsx` stays a server component; only sections that need hooks or events are client components.
- Shared types live in `types/index.ts`; a type used by one file stays in that file.
- `components/ui/` is only edited through the shadcn CLI or for small style tweaks.

8. Coding rules

These rules apply to every file, whether Salman or an AI assistant writes it.

Naming

- Components and their files: `PascalCase` (`Hero.tsx`).
- Hooks: `camelCase` starting with `use` (`useContact.ts`).
- Variables and functions: `camelCase`; constants: `UPPER_SNAKE_CASE` only for true constants.
- One exported component per file; file name matches the export.

TypeScript

- `strict` mode on; no `any`, no `@ts-ignore` without a comment explaining why.
- Type API responses with an interface and pass it to `axios.get<Type>()`.
- Props get an explicit type or interface.

React and Next.js

- Function components only; no class components.
- Do not fetch data in `useEffect`; use TanStack Query.
- Use `next/image` for images and `next/link` for internal routes.
- Add `"use client"` only to files that use state, effects, browser APIs or event handlers.
- Lists need a stable `key` (an id, never the array index when order can change).

Styling

- Tailwind utility classes only; no inline `style` and no extra CSS files, except theme variables in `globals.css`.
- Use theme tokens so dark mode works automatically.
- Merge conditional classes with the `cn()` helper.
- Layout containers use `max-w-6xl` with `px-4` for consistent width.

Quality

- ESLint and Prettier must pass before every commit.
- No `console.log` left in committed code (the contact route's placeholder log is removed once email works).
- No dead code, no commented-out blocks.
- Commit messages: short, present tense, one change each (`add projects section`).
- Branches: `main` is always deployable; work on `feat/<name>` branches.

9. UI/UX, accessibility, SEO and performance

The design is clean and minimal, mobile first, with Hero and Projects getting the most polish.

Design rules

- One primary accent color (the shadcn `primary` token); everything else stays neutral.
- One font family (Geist), with a clear size scale: `text-4xl` to `text-6xl` for the Hero heading, `text-3xl` for section titles, base size for body.
- Section spacing: `py-20`; card grids use `gap-6`.
- Breakpoints: design for 375 px first, then `sm` 640, `md` 768, `lg` 1024.
- Subtle motion only (hover and theme transitions); respect `prefers-reduced-motion`.
- Dark and light themes must both look finished; test every section in each.

Accessibility (target WCAG AA)

- Text contrast at least 4.5 : 1 in both themes.
- Every icon-only button or link has an `aria-label`.

- Every image has meaningful `alt` text; decorative images use `alt=""`.
- Form fields have visible labels and error messages.
- Full keyboard navigation with a visible focus ring; one `h1` per page, then `h2` per section in order.

SEO

- Title: `Salman Siddiq Shovo | Frontend Developer`; unique meta description under 160 characters.
- Open Graph and Twitter card tags with `og.png`.
- Canonical URL, `sitemap.xml`, `robots.txt` and a favicon.
- Optional JSON-LD `Person` schema with name, URL and social profiles.

Performance budget

Item	Limit
Largest Contentful Paint (mobile)	under 2.5 s
Cumulative Layout Shift	under 0.1
Total JavaScript on first load	under 200 KB gzipped
Any single image	under 300 KB, served as WebP or AVIF through `next/image`

- Load fonts with `next/font` so there is no layout jump.
- Lazy-load below-the-fold images and heavy client components.

10. Security, testing and deployment

The site holds no user data, so the main risks are leaked keys, spam through the contact form and broken pages after a deploy.

Security rules

- Secrets live only in `.env.local` and the hosting dashboard; `.env*` is in `.gitignore`.
- Only variables starting with `NEXT_PUBLIC_` may be read in client code, and none of them is a secret.
- Validate all input on the server with zod, even if the client already validated it.
- Never render user text as raw HTML (no `dangerouslySetInnerHTML` with form input).
- Do not publish a personal phone number or home address; email and social links are enough.

- Run `npm audit` before each release.

Testing

- Before each release, run `npm run lint`, `npx tsc --noEmit` and `npm run build`; all three must pass.
- Manually test on a phone-sized viewport, a tablet and a desktop in both themes.
- Test the Projects section three ways: normal, GitHub API blocked (error state) and a user with no repos (empty state).
- Test the contact form with valid data, an invalid email and a short message.
- Optional: add Vitest for the zod schema and Playwright for one end-to-end contact flow.

Deployment (Vercel)

1. Push the repo to GitHub.
2. Import it in Vercel and add the environment variables.
3. Deploy `main`; every pull request gets a preview URL.
4. Add a custom domain if wanted and set `NEXT_PUBLIC_SITE_URL` to it.
5. Run Lighthouse on the live URL and fix anything under 90.

Launch checklist

- [ ] Real email, CV, photo and OG image added.
- [ ] Contact route sends a real email (test from the live site).
- [ ] All social links checked.
- [ ] Metadata, sitemap and favicon present.
- [ ] Lighthouse 90+ in all four categories on mobile.
- [ ] No `console.log`, no TODO comments left in the code.

11. Milestones and open items

Build in five phases, finishing each one before starting the next; the starter code covers most of phases 1 to 3.

Phase 1: Setup

- [ ] Create the Next.js app and run the shadcn init.
- [ ] Install all packages and add the shadcn components.
- [ ] Add `QueryProvider`, `ThemeProvider` and `Toaster` in `layout.tsx`.
- [ ] Create `lib/axios.ts`, `types/index.ts` and `data/profile.ts`.

Phase 2: Core sections

- [ ] Navbar, Hero, About, Skills, Education, Footer.
- [ ] Theme toggle working in every section.

Phase 3: Dynamic features

- [ ] Projects section with `useGithubRepos` and its loading, error and empty states.
- [ ] Contact form with `useContact` and the `/api/contact` route.
- [ ] Real email sending connected.

Phase 4: Polish

- [ ] Replace placeholder text and add photo, CV and OG image.
- [ ] SEO metadata, sitemap, robots and favicon.
- [ ] Accessibility and responsive pass on every section.
- [ ] Optional: subtle animations with framer-motion.

Phase 5: Launch

- [ ] Run the launch checklist in section 10.
- [ ] Deploy to Vercel and test the live site.
- [ ] Add the live URL to the GitHub and LinkedIn profiles.

Open items for Salman

- Public email address to show on the site.
- DIU start year for the Education card.
- Final list of skills he is confident in.
- Any internship, job or freelance work for the Experience section.
- Whether to show the Facebook and Instagram links, or keep only LinkedIn and GitHub for a professional look.

Rules for AI assistants working on this repo

- Read this document before changing code, and follow sections 6 to 9 exactly.
- Change only what was asked; do not refactor unrelated files.
- Never add a library that is not in section 3 without asking first.
- Never invent personal facts; use the values in section 4 or leave a clearly marked TODO.
- After each change, confirm that lint, type check and build still pass.